

7 Cryptography

7.4 Basic Facts About Modular Arithmetic

Theorem 7.4.1 (*Basic facts about modular arithmetic*) when n is a natural number, the following are true:

1. $(a + b) \bmod n = (a \bmod n) + (b \bmod n)$
2. $(ab) \bmod n = (a \bmod n)(b \bmod n)$
3. $a^b \bmod n = (a \bmod n)^b \bmod n$

Example 7.4.2 Compute the following:

1. $128 \cdot 8935 \bmod 5$
2. $4356^{2532} \bmod 4355$
3. $4354^{2532} \bmod 4355$

Example 7.4.3 Compute the following:

1. $47^8 \bmod 49$
2. $65^5 \bmod 35$

Example 7.4.4 Your turn! Compute the following:

1. $12^8 \bmod 4$
2. $12^8 \bmod 5$
3. $5^{12} \bmod 4$
4. $98^{123} \bmod 99$

What else can we do with modular arithmetic?

Example 7.4.5 Compute each of the following:

1. $45^2 \bmod 20$
2. $(-35)^2 \bmod 50$

Notice that these give the same answer. It turns out that any computation that we do involving 45, can instead be done with the number -35, and vice versa. These two numbers are said to be equivalent mod 20. What other numbers are equivalent to -35 and 45?